

SPNN-512

a pseudo-invertible codec for latent diffusion

restoration · iterative editing · QR codes · visual anagrams

checkpoint step 49,000 (epoch 9)

DDNM restoration — SD-VAE vs SPNN

5,000 test images (85.8% LAION / 14.2% COCO) · FID against the originals of the same images · identical seed and noise per image · SPNN = step 49,000

Box inpainting — 26% missing, T=100

codec	LPIPS ↓	SSIM ↑	FID ↓	PSNR ↑	sec / image ↓
SD-VAE	0.0931	0.9005	4.27	24.35	9.99
SPNN	0.1558	0.8858	10.90	22.93	1.94

Box inpainting — 26% missing, T=200

codec	LPIPS ↓	SSIM ↑	FID ↓	PSNR ↑	sec / image ↓
SD-VAE	0.0947	0.9020	4.42	24.56	20.01
SPNN	0.1625	0.8854	12.04	22.98	3.89

Scattered inpainting — 66% missing, T=100

codec	LPIPS ↓	SSIM ↑	FID ↓	PSNR ↑	sec / image ↓
SD-VAE	0.4835	0.6659	63.17	15.84	9.99
SPNN	0.4654	0.6631	54.83	15.79	1.94

Scattered inpainting — 66% missing, T=200

codec	LPIPS ↓	SSIM ↑	FID ↓	PSNR ↑	sec / image ↓
SD-VAE	0.4789	0.6700	62.03	16.05	20.04
SPNN	0.4528	0.6697	52.77	16.08	3.88

Colorization — T=20

codec	LPIPS ↓	SSIM ↑	FID ↓	PSNR ↑	sec / image ↓
SD-VAE	0.3676	0.7056	25.42	14.23	2.00
SPNN	0.2871	0.8181	20.58	17.23	0.39

DDNM restoration — examples

DDNM restoration — one example per task

SD-VAE vs SPNN final (step 49k); same image, same seed and noise for both codecs



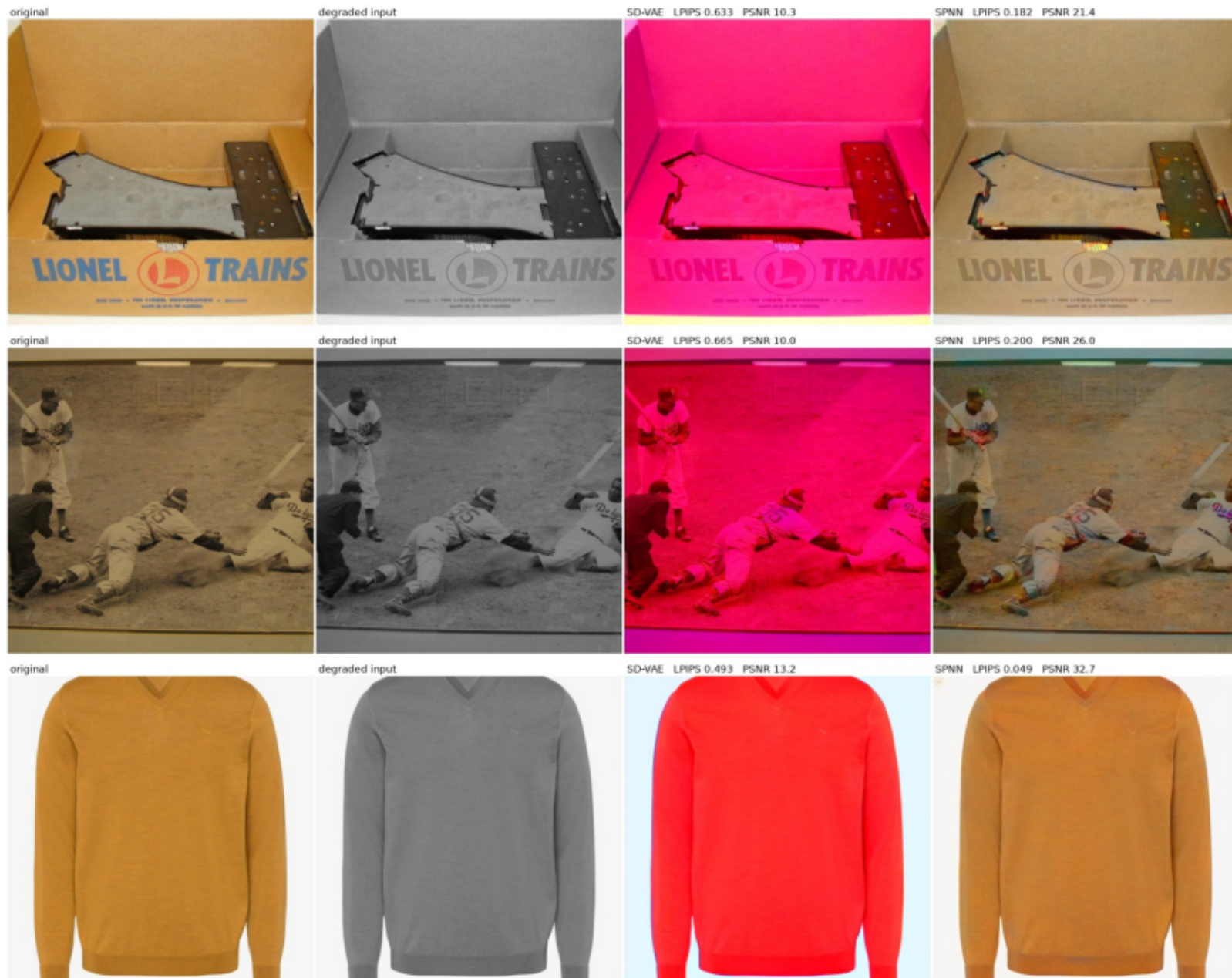
Scattered inpainting 66%, T=200

images where SPNN wins



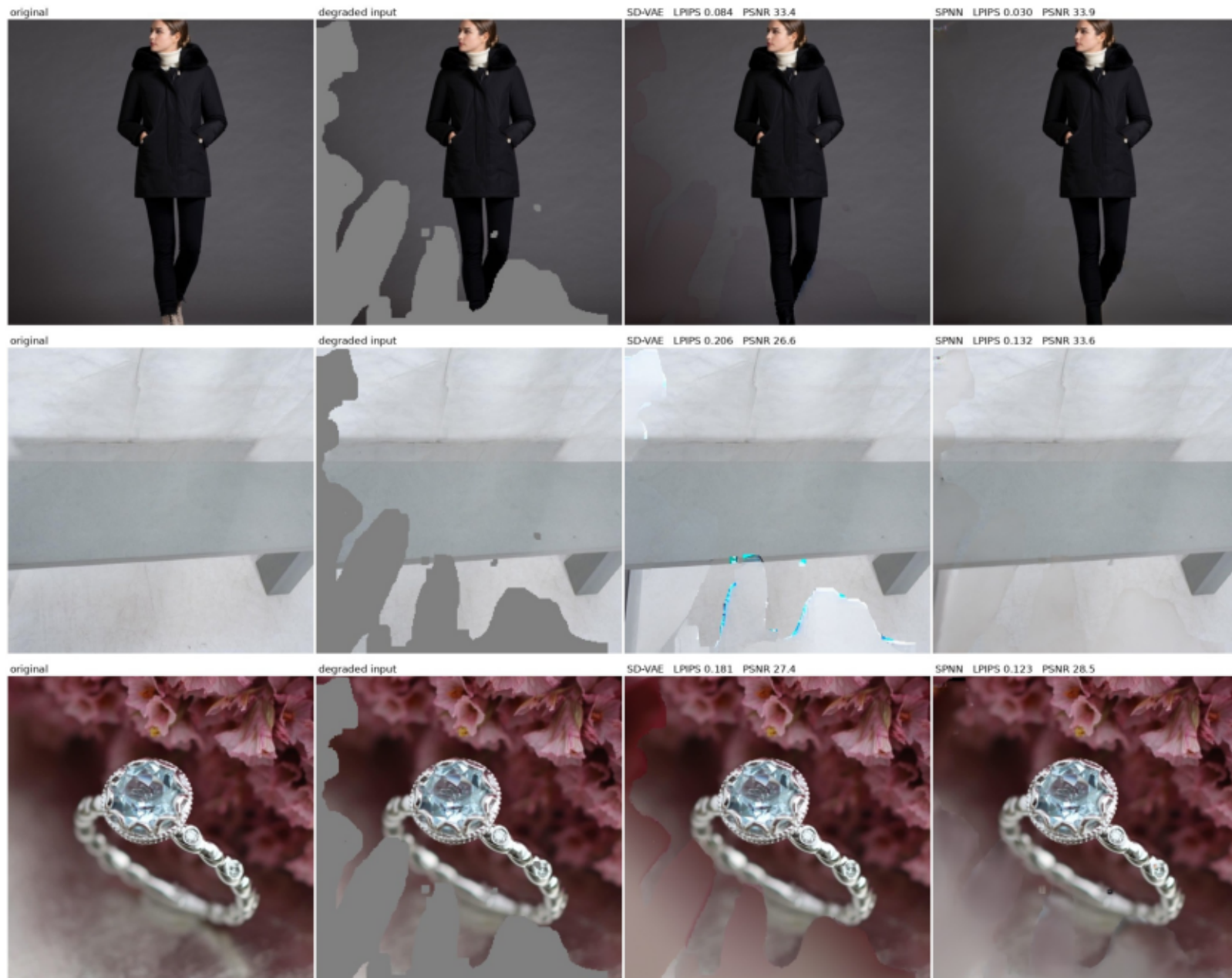
Colorization, T=20

images where SPNN wins



Box inpainting 26%, T=200

images where SPNN wins



Ablations and other tasks

Codec checkpoint — Paint-by-Example, 2 samples, scored against x¹

codec	MSE ↓ 5	MSE ↓ 15	MSE ↓ 25	LPIPS ↓ 5	LPIPS ↓ 15	LPIPS ↓ 25	SSIM ↑ 5	SSIM ↑ 15	SSIM ↑ 25	PSNR ↑ 5	PSNR ↑ 15	PSNR ↑ 25
SD-1.5 VAE	.007	.028	.046	.297	.683	.784	.684	.321	.234	21.76	15.56	13.48
SPNN epoch 3 (step 16,473, MSE only)	.028	.053	.057	.296	.719	.766	.646	.279	.294	21.48	12.83	12.50
SPNN final (step 49,000, +LPIPS)	.005	.024	.042	.121	.468	.628	.870	.606	.383	23.75	17.18	14.76

DiffQRCoder — 12 prompts, WeChat decoder

variant	scan rate	module error	CLIP	sec / image
sandwich · SD-VAE	1 / 12	0.674 %	0.274	5.81
sandwich · SPNN	2 / 12	0.585 %	0.277	2.63
sandwich · SD-VAE + MPGD	0 / 12	0.178 %	0.216	2.30
sandwich · SPNN + MPGD	4 / 12	0.000 %	0.278	0.71
gradient · SD-VAE + MPGD	12 / 12	0.030 %	0.212	3.15
gradient · SPNN + MPGD	11 / 12	0.000 %	0.209	0.91

Visual anagrams — 4 examples x 2 seeds

arm	alignment A ↑	concealment C ↑	sec / sample
SD1.5, latent-space views	0.2828	0.8079	2.20
SD1.5, decode-view-encode	0.2529	0.8187	15.75
SPNN, decode-view-encode	0.2276	0.6760	4.70

REED-VAE Table 1 — hybrid

179 ImagenHub samples · 25 alternating edits · scored against x¹ (Appendix A) · corrected ImagenHub alpha masks · best of the four ours/REED rows in bold

IP2P [BHE23]

method	MSE ↓ 5	MSE ↓ 15	MSE ↓ 25	LPIPS ↓ 5	LPIPS ↓ 15	LPIPS ↓ 25	SSIM ↑ 5	SSIM ↑ 15	SSIM ↑ 25	FID ↓ 5	FID ↓ 15	FID ↓ 25	PSNR ↑ 5	PSNR ↑ 15	PSNR ↑ 25
IP2P (paper)	0.020	0.110	0.150	0.33	0.69	0.76	0.60	0.23	0.18	105.80	247.00	271.70	17.78	10.15	8.59
+ REED (paper)	0.020	0.060	0.090	0.18	0.45	0.58	0.80	0.53	0.41	62.80	138.90	188.00	19.81	13.36	11.36
SD-1.5 VAE (ours)	0.040	0.114	0.150	0.41	0.69	0.74	0.54	0.25	0.21	134.83	216.18	236.18	15.52	9.85	8.54
+ SPNN (ours)	0.031	0.089	0.126	0.31	0.59	0.69	0.69	0.36	0.26	108.28	192.95	210.94	17.10	11.28	9.41
+ SD-VAE enc / SPNN dec (ours)	0.033	0.089	0.118	0.34	0.60	0.67	0.65	0.36	0.28	118.95	198.01	212.40	16.72	11.25	9.74

MagicBrush [ZMC*23]

method	MSE ↓ 5	MSE ↓ 15	MSE ↓ 25	LPIPS ↓ 5	LPIPS ↓ 15	LPIPS ↓ 25	SSIM ↑ 5	SSIM ↑ 15	SSIM ↑ 25	FID ↓ 5	FID ↓ 15	FID ↓ 25	PSNR ↑ 5	PSNR ↑ 15	PSNR ↑ 25
MagicBrush (paper)	0.020	0.080	0.140	0.31	0.71	0.80	0.65	0.21	0.13	103.50	267.00	295.80	18.84	11.35	8.75
+ REED (paper)	0.010	0.030	0.050	0.19	0.51	0.69	0.81	0.60	0.45	74.70	174.70	223.80	21.53	16.66	14.09
SD-1.5 VAE (ours)	0.035	0.108	0.157	0.38	0.73	0.79	0.57	0.18	0.12	120.83	255.71	273.77	16.62	10.09	8.21
+ SPNN (ours)	0.031	0.087	0.144	0.29	0.66	0.75	0.69	0.24	0.15	103.10	253.47	280.22	17.86	11.19	8.70
+ SD-VAE enc / SPNN dec (ours)	0.031	0.067	0.099	0.30	0.60	0.69	0.68	0.32	0.20	110.53	217.46	242.04	18.08	12.88	10.53

DiffEdit [CVSC22]

method	MSE ↓ 5	MSE ↓ 15	MSE ↓ 25	LPIPS ↓ 5	LPIPS ↓ 15	LPIPS ↓ 25	SSIM ↑ 5	SSIM ↑ 15	SSIM ↑ 25	FID ↓ 5	FID ↓ 15	FID ↓ 25	PSNR ↑ 5	PSNR ↑ 15	PSNR ↑ 25
DiffEdit (paper)	0.030	0.060	0.080	0.34	0.62	0.73	0.65	0.36	0.21	160.70	246.30	301.90	15.99	12.59	11.21
+ REED (paper)	0.030	0.070	0.080	0.30	0.55	0.68	0.69	0.48	0.40	160.30	226.90	246.50	16.24	12.91	11.44
SD-1.5 VAE (ours)	0.026	0.073	0.106	0.34	0.69	0.77	0.63	0.25	0.16	152.61	255.64	273.73	16.26	11.48	9.84
+ SPNN (ours)	0.024	0.066	0.105	0.26	0.58	0.71	0.73	0.38	0.19	152.81	254.20	322.40	16.71	11.96	9.96
+ SD-VAE enc / SPNN dec (ours)	0.021	0.051	0.069	0.27	0.55	0.65	0.72	0.44	0.31	148.62	233.54	250.93	17.25	13.05	11.73

REED-VAE Table 1 — hybrid

179 ImagenHub samples · 25 alternating edits · scored against x¹ (Appendix A) · corrected ImagenHub alpha masks · best of the four ours/REED rows in bold

PbE [YGZ*23]

method	MSE ↓ 5	MSE ↓ 15	MSE ↓ 25	LPIPS ↓ 5	LPIPS ↓ 15	LPIPS ↓ 25	SSIM ↑ 5	SSIM ↑ 15	SSIM ↑ 25	FID ↓ 5	FID ↓ 15	FID ↓ 25	PSNR ↑ 5	PSNR ↑ 15	PSNR ↑ 25
PbE (paper)	0.020	0.040	0.070	0.26	0.60	0.71	0.66	0.33	0.22	83.50	209.30	253.60	18.55	13.88	11.74
+ REED (paper)	0.020	0.030	0.040	0.20	0.44	0.59	0.77	0.61	0.54	74.20	141.10	178.30	19.62	16.19	14.43
SD-1.5 VAE (ours)	0.019	0.044	0.069	0.26	0.60	0.71	0.67	0.33	0.22	81.61	207.10	246.94	18.70	13.91	11.80
+ SPNN (ours)	0.019	0.043	0.067	0.24	0.52	0.65	0.73	0.46	0.30	91.02	216.14	294.72	18.93	14.46	12.19
+ SD-VAE enc / SPNN dec (ours)	0.016	0.027	0.039	0.21	0.47	0.59	0.76	0.52	0.38	76.22	177.61	220.44	20.00	16.57	14.62

SD Inpainting [RBL*22]

method	MSE ↓ 5	MSE ↓ 15	MSE ↓ 25	LPIPS ↓ 5	LPIPS ↓ 15	LPIPS ↓ 25	SSIM ↑ 5	SSIM ↑ 15	SSIM ↑ 25	FID ↓ 5	FID ↓ 15	FID ↓ 25	PSNR ↑ 5	PSNR ↑ 15	PSNR ↑ 25
SD Inpainting (paper)	0.010	0.060	0.110	0.29	0.69	0.78	0.67	0.22	0.14	95.10	255.80	283.40	20.46	12.31	9.72
+ REED (paper)	0.010	0.030	0.050	0.17	0.47	0.65	0.80	0.57	0.41	72.70	166.10	210.40	23.14	16.66	13.61
SD-1.5 VAE (ours)	0.017	0.045	0.082	0.26	0.62	0.73	0.67	0.30	0.18	85.17	223.61	274.95	19.27	13.72	10.96
+ SPNN (ours)	0.021	0.047	0.077	0.26	0.55	0.68	0.72	0.43	0.24	107.06	247.32	320.02	18.39	13.93	11.48
+ SD-VAE enc / SPNN dec (ours)	0.014	0.024	0.037	0.21	0.48	0.60	0.76	0.51	0.36	81.63	189.31	230.83	20.85	16.91	14.55

Paint-by-Example — three codecs

SD-1.5 VAE / SPNN / SD-VAE encoder + SPNN decoder

PbE — 25 alternating edits · three codecs · scored against x^*

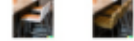
conditioning alternates every iteration → PbE takes no text prompt, only a reference crop

291614_32

odd steps = x^*

even steps = x^*

brushed region of the target / of the source; local caption: "kitchen with cat sitting on metal chair."



148240_32

odd steps = x^*

even steps = x^*

brushed region of the target / of the source; local caption: "man holding banana"



100081_35

odd steps = x^*

even steps = x^*

brushed region of the target / of the source; local caption: "two people in an ocean playing with a yellow ball"



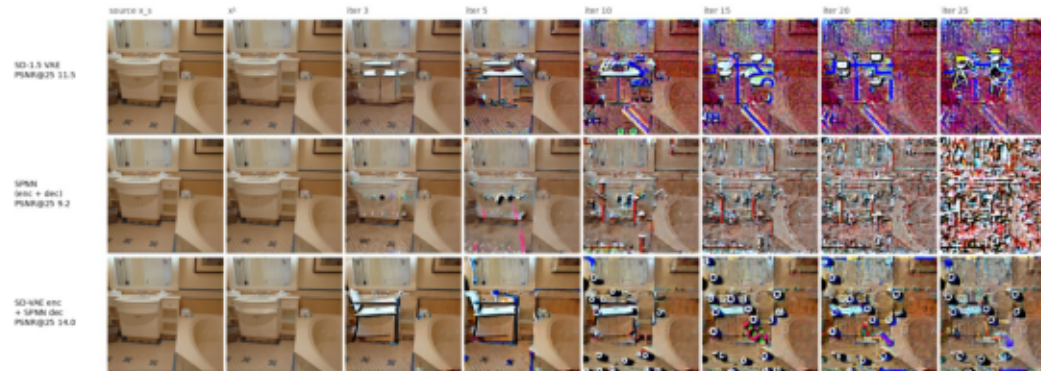
DiffEdit — three codecs

SD-1.5 VAE / SPNN / SD-VAE encoder + SPNN decoder

DiffEdit — 25 alternating edits · three codecs · scored against x¹

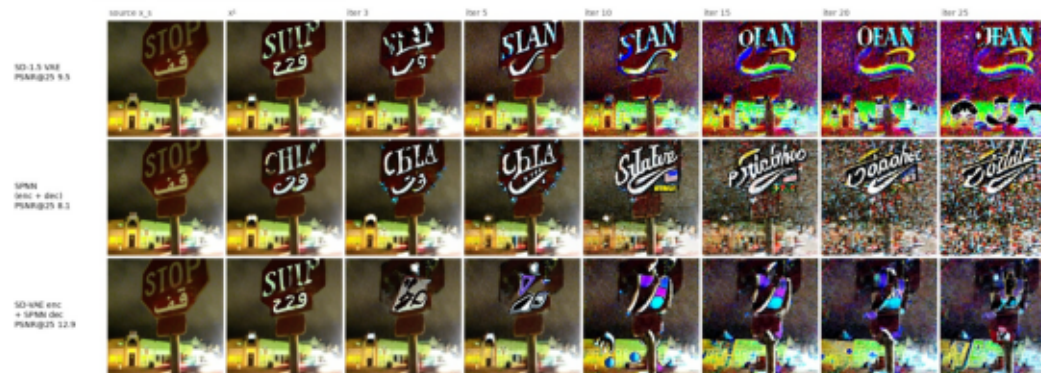
conditioning alternates every iteration.
109093_11

odd steps → "A relaxing oasis: this bathroom features a large jacuzzi tub filled with water and adorned with"
even steps → "This bathroom has a large jacuzzi tub and a large mirror"



24629_11

odd steps → "A red stop sign with Chinese writing sitting on the side of a street."
even steps → "A red stop sign sitting on the side of a street."



201614_12

odd steps → "A kitchen with a cat sitting on a large counter lined with metal chairs."
even steps → "A kitchen with a cat sitting on a large counter lined with cloth covered metal chairs."



SD Inpainting — three codecs

SDInpaint — 25 alternating edits (108871_t2)

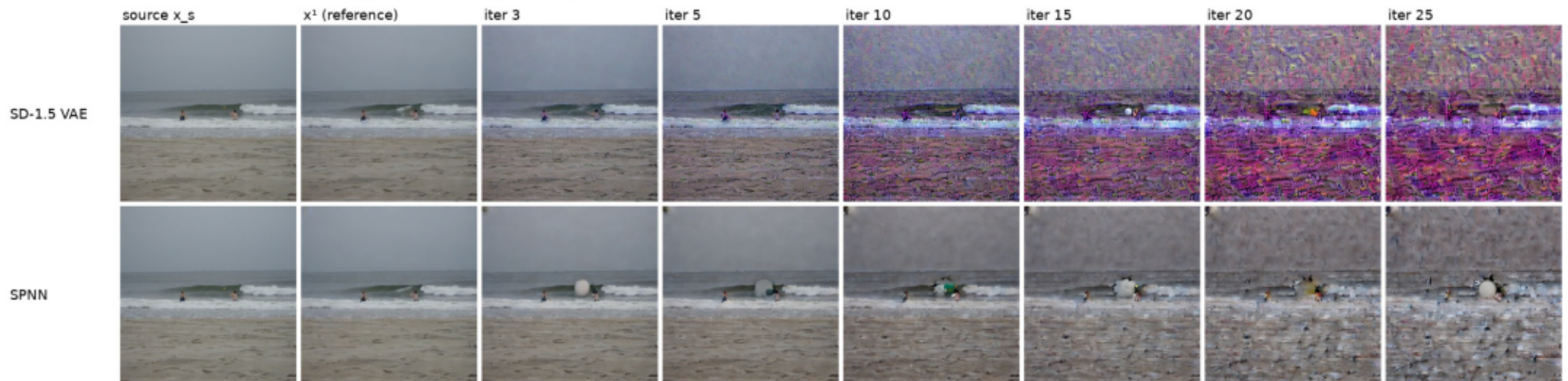
plastic pool with zebra and lion · scored against x^1



Paint-by-Example — 25 alternating edits

Paint-by-Example — 25 alternating edits (100081_t1)

"change the frisbee into a ball" reversed and re-applied each iteration; scored against x^1

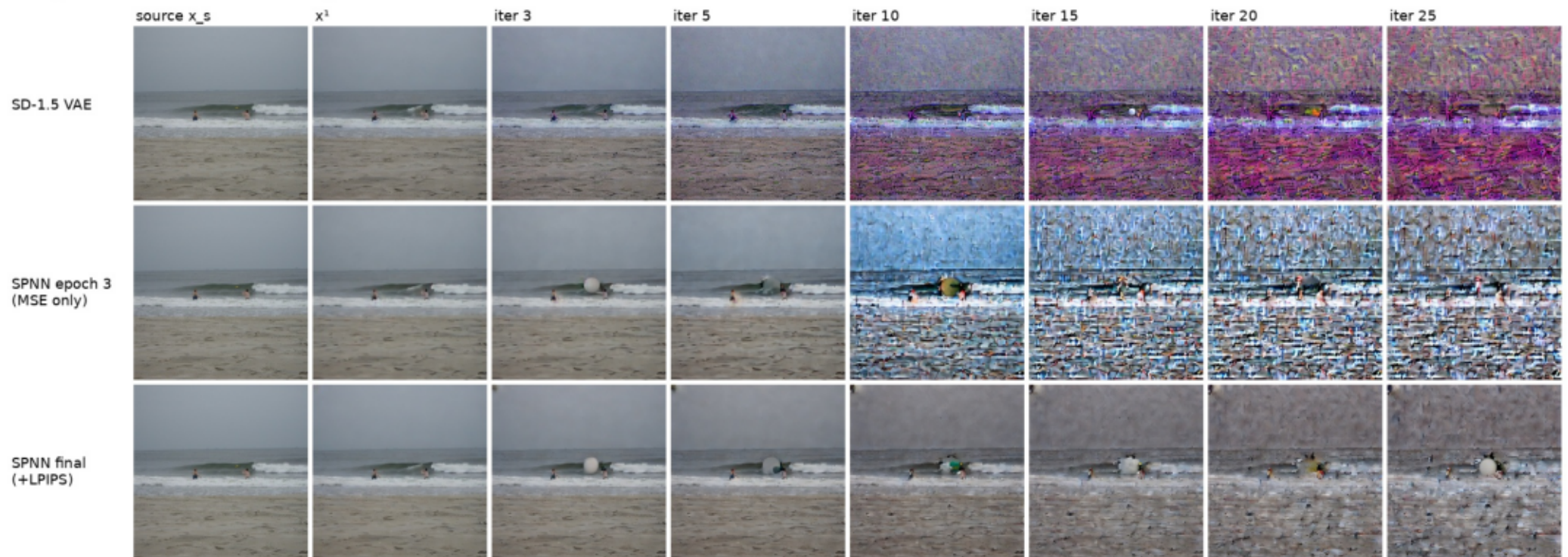


Codec checkpoint ablation

same architecture, LPIPS loss added at step 21,783

Paint-by-Example, 25 alternating edits — codec checkpoints (100081_t1)

"change the frisbee into a ball" · scored against x^1

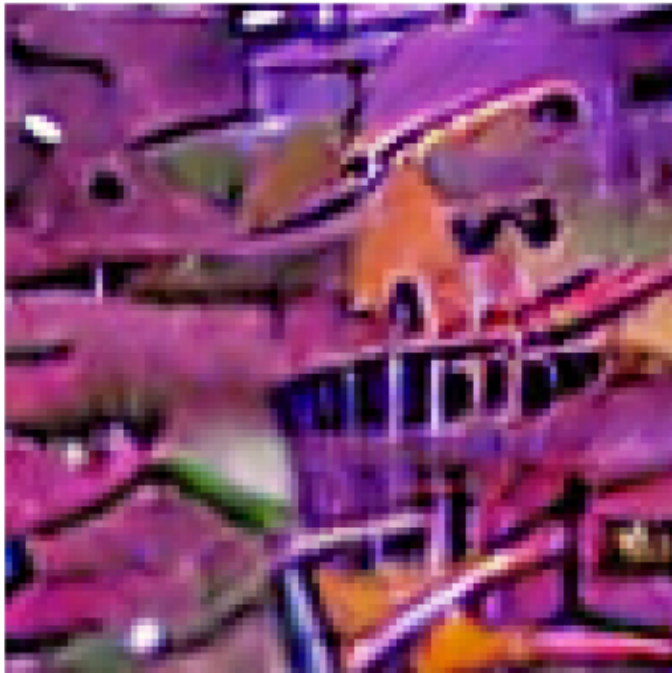


The 8-pixel latent-grid artifact

Iteration 25, 96x96 px crop at 4x — the artifact texture

same region of the same image; nearest-neighbour zoom so no resampling is added

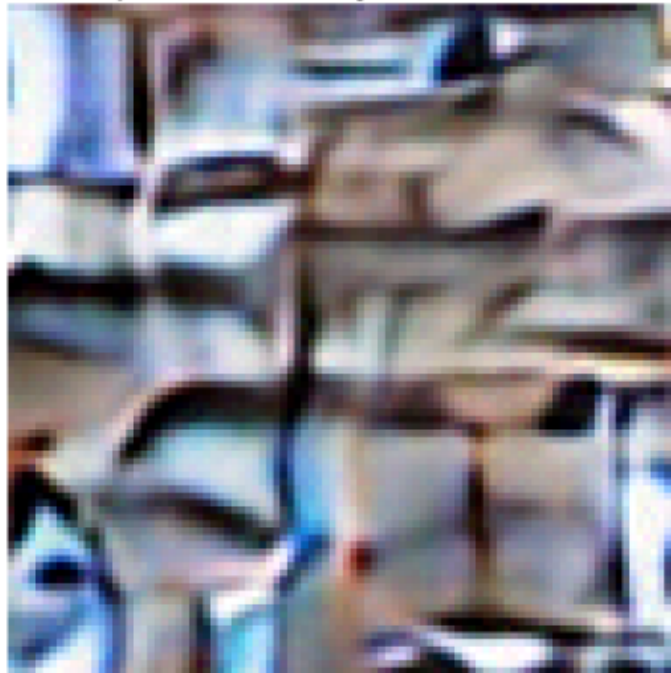
SD-1.5 VAE



mean 8x8 cell error (range 19e-3)



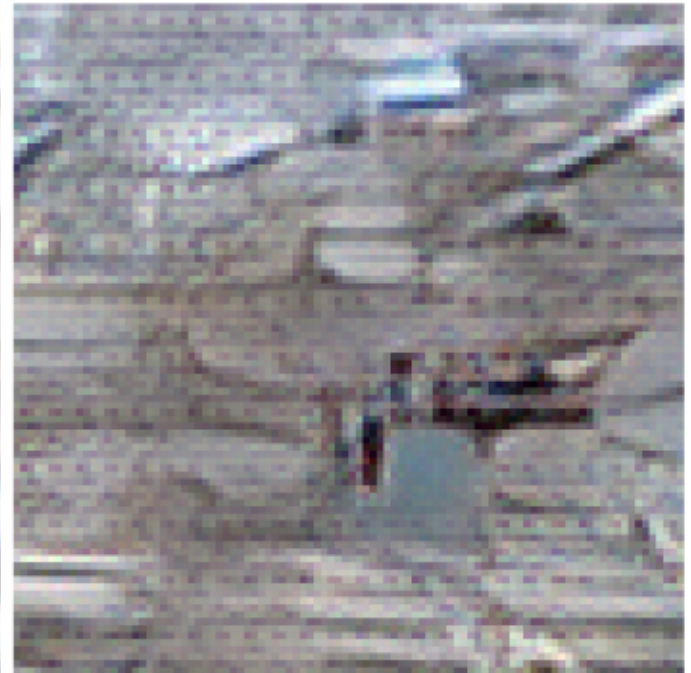
SPNN epoch 3 (MSE only)



mean 8x8 cell error (range 48e-3)



SPNN final (+LPIPS)



mean 8x8 cell error (range 84e-3)



DiffQRCoder — sandwich + MPGD

green = scans (WeChat)



DiffQRCoder — sandwich + MPGD

green = scans (WeChat)



Visual anagrams — flip

flip_campfire_man

identity: "an oil painting of people around a campfire" | flip: "an oil painting of an old man"

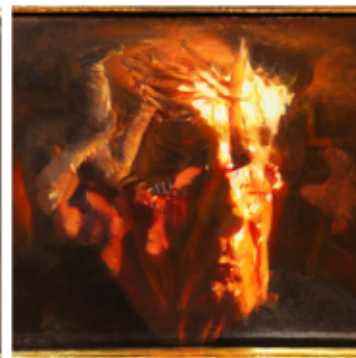
seed 0
identity

seed 0
flip

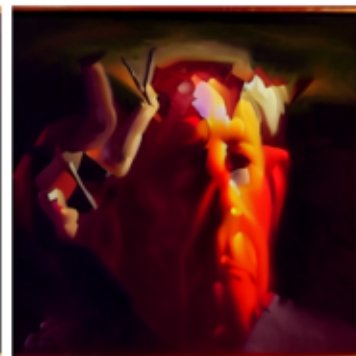
seed 1
identity

seed 1
flip

sd15-latent



sd15-sandwich



spnn-sandwich



Visual anagrams — negate

negate_landscape_houseplants

identity: "a lithograph of a landscape" | negate: "a lithograph of houseplants"

seed 0
identity

seed 0
negate

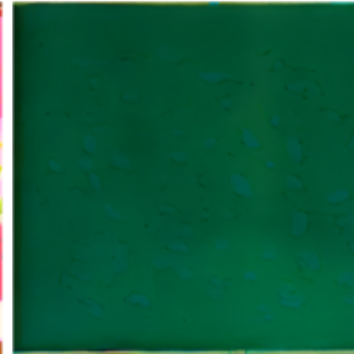
seed 1
identity

seed 1
negate

sd15-latent



sd15-sandwich



spnn-sandwich

